

# Nikunj Pradhan

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## EDUCATION

**Missouri University of Science and Technology**  
*Bachelor of Science, Computer Science, Focus: Data Science*

GPA: 4.0/4.0  
*Aug 2023 - Expected Dec 2026*

## EXPERIENCE

### Data Science Co-Op

Jan 2025 – Aug 2025

*Hunter Engineering Company*

*Bridgeton, MO*

- Elevated deep learning model workflow accuracy from 49% to 84% through hyperparameter optimization, active learning, and selective data clustering, improving model reliability for diverse usage.
- Reduced model training time by 66% by migrating YOLO detection/classification models to PyTorch Lightning with Distributed Data Parallel for multi-GPU training.
- Delivered full-stack apps (FastAPI, React, SQL Server) production model dashboards, dataset analytical applications, and AI-assisted annotation.
- Engineered geometric AI perspective-correction model, increasing classification accuracy by 35% and reducing manual data-cleaning effort.
- Automated cross-cloud benchmarking pipelines across AWS and Azure, enabling direct performance comparisons of vendor AI/ML models to drive strategic evaluation and decision-making.

### Computer Vision Researcher

Aug 2023 – Dec 2024

*Missouri University of Science and Technology*

*Rolla, MO*

- Optimized algorithms, increasing blob analysis accuracy 62% through hyperparameter tuning, and algorithmic refinement using C++ OpenCV and MATLAB.
- Designed a C++/MATLAB application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling with Point Cloud Library.
- Utilized advanced computer vision techniques, including watershed segmentation and 3D model triangulation via ray tracing, for improved object detection and modeling.
- Led weekly research coordination sessions to discuss algorithmic approaches, share updates, conduct code reviews, and plan future experiments.

## PROJECTS

### Nepali Devanagari OCR Pipeline | *PyTorch, Python, FastAPI*

- Designed and implemented a text detection + CRNN workflow covering 70+ Nepali consonants, vowels, punctuation marks, and matras, achieving 82% accuracy in real-world OCR applications.
- Developed an encoder-decoder transformer OCR model, attaining 92% paragraph-level recognition accuracy and enhancing robust image-to-text processing capabilities.
- Collected and annotated 400+ Nepali text images, applying augmentations to enhance training robustness.
- Evaluated models with Word/Character Error Rate metrics and error analysis to identify class-specific weaknesses and guide refinements.
- Implemented FastAPI inference service to deploy models for batch and real-time text recognition pipelines.

### Himalayan Hut | *FastAPI, React, PostgreSQL, MinIO, Scikit-learn*

- Developed a full-stack movie watchlist app with personalized recommendations, allowing users to create, track, rate, and share watchlists.
- Implemented hybrid recommendation system combining content-based filtering and collaborative filtering, using per-user caching for fast, accurate suggestions.
- Deployed scalable web application using PostgreSQL, MinIO, Kubernetes (k3s), and NGINX, integrating secure email-based 2FA login.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, JavaScript, SQL, MATLAB, Bash, HTML/CSS

**Machine Learning & AI:** Pytorch, Scikit-learn, ONNX, CNN/CRNN, Recommendation Systems, Transformers

**Databases:** MongoDB, PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO

**Data Science Tools:** Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV

**Web Development & APIs:** FastAPI, React, Node.js, Express.js, WebSockets

**Cloud & DevOps:** AWS (S3, EC2, Lambda), Azure ML, Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu